

Body Landmark Detection & Measurement Analysis

Objective

The objective of today's task was to test body landmark detection using MediaPipe Pose and analyze how body measurements can improve the accuracy of a Virtual Try-On system.

Approach Followed

Initially, the task suggested testing landmark detection using a webcam. However, for better experimentation, reproducibility, and documentation, I implemented the system using uploaded images instead of a live camera feed.

This approach allowed me to test different body poses under controlled conditions and easily capture screenshots for analysis. Using image uploads also ensured consistent inputs and simplified the evaluation process.

Tools Used

- Python
- OpenCV
- MediaPipe Pose
- Static Image Upload Method

```
1  import cv2
2  import mediapipe as mp
3  mp_pose = mp.solutions.pose
4  pose = mp_pose.Pose(static_image_mode=True)
5  mp_draw = mp.solutions.drawing_utils
6  image_path = "person_d.jpg"
7  image = cv2.imread(image_path)
8  rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
9  results = pose.process(rgb)
10 if results.pose_landmarks:
11     mp_draw.draw_landmarks(
12         image,
13         results.pose_landmarks,
14         mp_pose.POSE_CONNECTIONS
15     )
16 cv2.imshow("Landmark Detection", image)
17 cv2.waitKey(0)
18 cv2.destroyAllWindows()
```

Experiment 1: Standing Straight Pose

Input

A full-body image of a person standing straight facing the camera.

Process

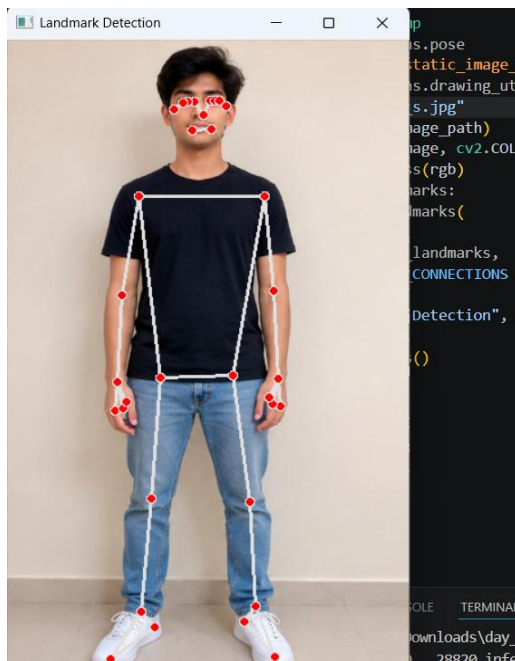
The uploaded image was processed using MediaPipe Pose. The model detected key body landmarks such as:

- Head
- Shoulders
- Elbows
- Wrists
- Hips
- Knees
- Ankles



Output

The pose estimation model successfully identified body landmarks and generated a complete skeletal structure.



Observation

The standing pose provided the highest accuracy because all major body parts were clearly visible. Landmark positions were stable and correctly aligned with the body.

Experiment 2: Hands Raised Pose

Input

A full-body image of the same person with both hands raised above the head.

Process

The uploaded image was analyzed using the same MediaPipe Pose model.

Output

The model successfully tracked shoulder, elbow, wrist, and upper body landmarks while maintaining overall pose consistency.



Observation

MediaPipe accurately detected raised arms and preserved body structure. The pose estimation remained reliable even with significant arm movement.

Experiment 3: Side Pose

Input

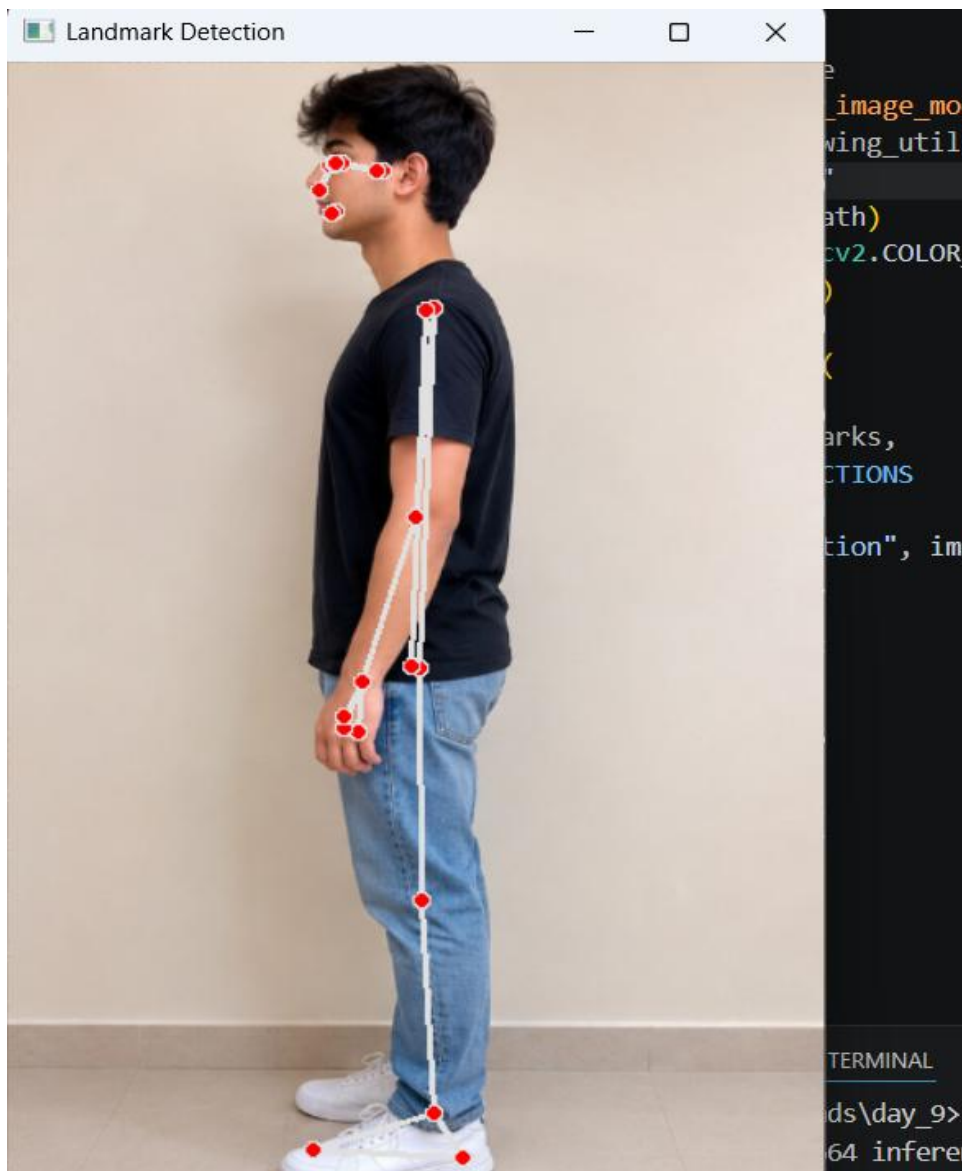
A side-view image of the same person.

Process

The image was processed through the pose detection pipeline.

Output

Most body landmarks were detected successfully despite partial visibility of some body parts.



Observation

The accuracy was slightly lower compared to frontal poses because certain landmarks were partially hidden. However, overall pose estimation remained satisfactory.

Landmark Detection Accuracy Analysis

Pose Type	Accuracy Level	Remarks
Standing Straight	Excellent	All landmarks clearly visible
Hands Raised	Excellent	Accurate arm and upper body detection
Side Pose	Good	Minor reduction due to occlusion

Importance of Body Measurements in Virtual Try-On

Body measurements play a critical role in improving virtual try-on realism and garment fitting.

Key Measurements

- Shoulder Width
- Chest Width
- Waist Width
- Hip Width
- Arm Length
- Leg Length
- Overall Height

Benefits

Better Garment Alignment

Body measurements help position garments accurately on the user's body.

Improved Scaling

Clothing can be resized according to the user's dimensions.

Realistic Visualization

Garments appear more natural and fit the body shape correctly.

Enhanced User Experience

Users receive a more accurate representation of how clothes would look in reality.

Results

- Successfully implemented MediaPipe Pose Detection using uploaded images.
- Tested landmark detection on three different body poses.
- Verified that MediaPipe can reliably detect body landmarks under different conditions.

- Observed highest accuracy in frontal poses.
- Studied the role of body measurements in improving virtual try-on performance.

Conclusion

Today's work demonstrated that MediaPipe Pose can effectively detect human body landmarks using uploaded images. Although the original task suggested webcam-based testing, the image-upload approach provided a more controlled and documentation-friendly environment.

The experiments showed that frontal poses achieve the highest detection accuracy, while side poses may experience slight reductions due to hidden body parts. Additionally, body measurements were identified as an important factor for improving garment fitting, scaling, and overall realism in Raritone's Virtual Try-On system.

Therefore, combining accurate landmark detection with body measurement estimation can significantly enhance the performance and user experience of the Raritone Virtual Try-On platform.